NEUROPLANT INDUSTRIAL INTELLIGENCE LABS

Standard Operating Procedure & Safety Compliance Mandate

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Target Unit:	Unit 3 - Hydrocarbon Processing	Primary Asset:	Centrifugal Pump P-204
Governing Standard:	OSHA 29 CFR 1910.147 (LOTO)	Secondary Code:	API 610 12th Edition
Associated Valves:	Suction V-101, Discharge V-102	Sealing Element:	Cartridge Mechanical Seal MS-204
Risk Level:	High (Hazardous Steam & Fluid)	Responsible Dept:	Mechanical Reliability & Safety

1. Scope and Operational Purpose

This standard establishes mandatory safety requirements for mechanical isolation, pressure bleed-down, and mechanical seal replacement for **Centrifugal Pump P-204** located in Unit 3. Operation involves high-pressure steam (400 psi) and volatile hydrocarbon feed at 220 deg C.

2. Step-by-Step Isolation Protocol (LOTO)

Step 2.1: Inform Unit Control Room operator before initiating isolation sequence.
Step 2.2: Manually close suction isolation valve **V-101** and tag out.
Step 2.3: Manually close discharge isolation valve **V-102** and lock with approved hasp.
Step 2.4: Verify complete depressurization on local pressure transmitter **PT-302**.
Step 2.5: Disengage power feed at breaker **MCC-4B** following OSHA Lockout/Tagout mandates.

3. Inspection & Compliance Verification

Inspect cartridge seal **MS-204** for face degradation. Replace seal if primary ring face thickness is below 1.2mm. In accordance with historical Incident **INC-2023-09**, under no circumstances should suction valve V-101 be cracked open during thermal flush without written authorization.